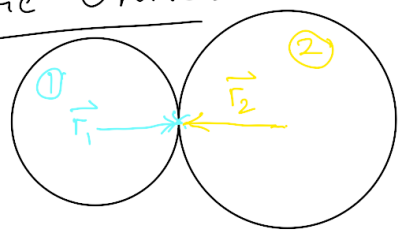
(1) The homogeneous Universe

(A) Static Universe



$$\vec{a}_1 = -\frac{GM(<r_1)}{r_1^3}\vec{r}_1 = -\frac{4\pi G\rho\,\vec{r}_1}{3}$$

$$\vec{a}_2 = -\frac{4\pi G\rho}{3}\vec{r}_2$$

$$\vec{a}_1 \neq \vec{a}_2$$

(B) If Universe is dynamically active, then observer 1 and observer 2 are in diff^t ref frames. So must account for acceleration of one obs. w.r.t. to the other.

$$\vec{a}_1 = -\frac{4\pi G\rho}{3}\vec{r}_1$$

$$\vec{a}_{12} = -\frac{4\pi G\rho}{3}(\vec{r}_1 - \vec{r}_2)$$

$$\vec{a}_1 - \vec{a}_{12} = -\frac{4\pi G\rho\,\vec{r}_2}{3}$$

So a Euclidean "linearly accelerating" Universe is self-consistent.

ⓒ Dynamical eq of motion of Universe

$$r(t) = a(t)\, \ell$$

Comoving coordinate . dimension of ℓ [m]

$a(t)$ is dimensionless function of time only.

expansion factor.

$$v(t) = \frac{dr}{dt} \equiv \dot{r}$$

$$v(t) = \dot{a}(t)\ell = \left(\frac{\dot{a}}{a}\right) r(t)$$

$$\vec{v}(t) = \left(\frac{\dot{a}}{a}\right) \vec{r}(t)$$

$$\vec{A}(t) = \frac{d\vec{v}}{dt} = \left(\frac{\ddot{a}}{a}\right) \vec{r}(t)$$

$\vec{a}_{12}$ " $(\vec{r}_1 - \vec{r}_2)$

$$-\frac{4\pi \rho G \vec{r}(t)}{3} = \frac{\ddot{a}}{a} \vec{r}(t)$$

$$\boxed{-\frac{4\pi \rho G}{3} = \frac{\ddot{a}}{a}}$$

$\times\ \dot{a}a$ and integrate assuming mass is conserved :

$\rho a^3 = \text{const}$

$$\boxed{\dot{a}^2 - \frac{8\pi G \rho a^2}{3} = 2E}$$

(D) General relativity formalism

$$ds^2 = g_{\mu\nu}\, dx^\mu dx^\nu$$

$$= (c\,dt \;\; dr \;\; d\theta \;\; d\phi) \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{a^2(t)}{1-kr^2} & 0 & 0 \\ 0 & 0 & a^2(t) r^2 & 0 \\ 0 & 0 & 0 & a^2(t) r^2 \sin^2\theta \end{pmatrix} \begin{pmatrix} c\,dt \\ dr \\ d\theta \\ d\phi \end{pmatrix}$$

$$= -c\,dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2 d\theta^2 + r^2 \sin^2\theta\, d\phi^2\right]$$

$$\left(\eta_{\mu\nu} = \eta^{\mu\nu} \text{ but } g^{\mu\nu} \neq g_{\mu\nu}\ !!\right)$$

This is called the Friedmann - Lemaître - Robertson - Walker metric (FLRW) using space-like convention $(-,+,+,+)$.

Universe is homogeneous & isotropic filled with a perfect fluid & we sit in the comoving frame in which the fluid is at rest:

$$T^\nu_{\ \mu} = \begin{pmatrix} -\rho(t)c^2 & & & \\ & P(t) & & \\ & & P(t) & \\ & & & P(t) \end{pmatrix}$$

energy - momentum tensor.

with $\rho(t)$ and $P(t)$ are (time dependent) average density & pressure of the fluid.

$$T_{\mu\alpha} = \left(\rho(t) + P(t)/c^2\right) U_\mu U_\alpha + P(t)\, g_{\mu\alpha}$$

where $U_\mu = (-c, 0, 0, 0)$ covariant 4-velocity of fluid at rest.

multiply by contravariant metric tensor

$$g^{\alpha\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{1-kr^2}{a^2} & 0 & 0 \\ 0 & 0 & 1/(a^2(t) r^2) & 0 \\ 0 & 0 & 0 & \frac{1}{a^2(t) r^2 \sin^2\theta} \end{pmatrix}$$

$$T^{\nu}_{\mu} = \left(\rho(t) + P(t)/c^2\right) U_\mu U^\nu + P(t)\, \delta_\mu^{\ \nu}$$

$$g^{\mu\nu} U_\mu U_\nu = U_\mu U^\mu = U_0 U^0 = -c^2$$

Specify the nature of fluid by an equation of state: $P \equiv P(\rho)$

Convention: $(+,-,-,-)$ ← metric sign; ← energy-momentum tensor; ← Riemann tensor

Einstein's field equation.

$$\boxed{R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = -\frac{8\pi G}{c^4} T_{\mu\nu} + \Lambda g_{\mu\nu}}$$

$R_{\mu\nu}$ and R are called Ricci tensor and scalar they are functions of $g_{\mu\nu}$ and 1st & 2nd der. w.r.t to coordinates. Λ is called the cosmological constant.

Using explicit form of FLRW metric & ideal fluid energy mom. tensor –

R_{00} →
$$\dot{a}^2 = \frac{8\pi G a^2}{3}\rho(t) + \frac{\Lambda a^2 c^2}{3} - kc^2$$

R_{rr} →
$$\ddot{a} = -\frac{4\pi G a}{3}\left(\rho(t) + 3\frac{P(t)}{c^2}\right) + \frac{\Lambda a(t) c^2}{3}$$

Friedmann – Lemaître (FL) equations –
+ EOS : closed system –

Hubble parameter : $H(t) = \frac{\dot{a}}{a}(t) \equiv$ expansion rate of Universe –

Critical density : $\rho_c(t) = \frac{3H^2(t)}{8\pi G}$

Cosmological redshift : $z \equiv \frac{a_0}{a(t)} - 1 \quad (a_0 = a(t=t_0)$ ↓ now.)

instead of $a \longrightarrow$ directly observable $z \equiv \frac{\lambda_0}{\lambda(t)} - 1$

rewrite 1st FL equation :

$$\rho_c(z) = \rho(z) + \frac{\Lambda c^2}{8\pi G} - \frac{3k c^2 (1+z)^2}{8\pi G}$$

$$\rho_X(z)/\rho_c(z) \equiv \Omega_X(z)$$

$$\boxed{\Omega_M(z) + \Omega_\gamma(z) + \Omega_V(z) + \Omega_K(z) = 1}$$

make scalings w. z explicit:

$$\rho_M(z) \propto (1+z)^3 \quad , \quad \rho_\gamma(z) \propto (1+z)^4$$

$$\rho_V(z) = c^t \quad . \quad \rho_K(z) \propto (1+z)^2$$

$$\boxed{H^2(z) = H_0^2\left(\Omega_{M,0}(1+z)^3 + \Omega_{\gamma,0}(1+z)^4 + \Omega_{V,0} + \Omega_{K,0}(1+z)^2\right)}$$

$$\Omega_{M,0} = 0.317 \quad \Omega_{V,0} = 0.683$$

$$\Omega_{K,0} = 0.0 \quad , \quad \Omega_{\gamma,0} = 9 \cdot 10^{-5}$$

$$H_0 = 67.2 \ km/s/Mpc \quad (Planck)$$

Conformal time : $\eta \qquad d\eta \equiv \frac{dt}{a(t)}$

$$\rightarrow ds^2 = a^2[\eta]\left(-c^2 d\eta^2 + \frac{dr^2}{1-Kr^2} + r^2 d\theta^2 + r^2 \sin^2\theta \, d\varphi^2\right)$$

Comoving, Particle Horizon

$$R_h(t) = c\eta = c\int_0^t \frac{dt'}{a(t')} = R_h(z) = c\int_z^\infty \frac{dz'}{H(z')}$$

Proper distance (physical) is equal to comoving distance $\times$ $a(t)$ $\longrightarrow$ means $R_{phys}(z=0) = R_{com}^{(z=0)}$ as $a(t_0) = 1$ by convention. $R_{phys}(z) = \frac{R_h(z)}{1+z}$

proper distance to particle horizon @ redshift z